



United International University

Department of Computer Science and Engineering

CSE 2213/CSI 219: Discrete Mathematics

Midterm Examination : Spring 2026

Total Marks: 30 Time: 1 Hour 30 Minutes

Answer all the 5 questions. Numbers to the right of the questions denote their marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. (a) Consider the following propositions: [1*2=2]

p : I wake up late.
 q : I miss the bus.
 r : I reach class on time.

Express the following statements using propositions and logical connectors:

- I reach class on time on all days when I do not wake up late.
- If I miss the bus, I do not reach class on time; otherwise I reach class on time.

- (b) Determine the contrapositive and converse of the following statement: [2]

“Unless you check the deadline, you assume there is still time to submit the assignment.”

2. (a) Using truth table, prove that the following expression is a contradiction: $(p \rightarrow q) \wedge (q \rightarrow \neg p) \wedge (p \vee q)$ [2]

- (b) Using logical equivalence laws, prove or disprove the following equivalence: [2]

$$\neg((\neg p \wedge q) \rightarrow (\neg r \vee s)) \equiv (\neg p \wedge q) \wedge (r \wedge \neg s)$$

3. (a) Consider the following predicates (the domain of the variables is all people in the world): [1 × 4 = 4]

$P(x)$: x is a soldier.
 $Q(x)$: x is brave.
 $R(x)$: x is calm.

Express the following statements using these predicates, appropriate quantifiers and logical connectives:

- Every soldier is brave and calm.
- There is a soldier who is not calm.
- Not every brave man is a soldier.
- No soldier is afraid.

- (b) With proper explanation, find out whether the following propositions are true or false (the domain of the variables is all real numbers): [1 × 4 = 4]

i. $\exists x (x^2 - 5x + 6 = 0)$

ii. $\forall x \left(\frac{x^2}{x} \geq 1 \right)$

iii. $\forall x \exists y \left(\frac{x}{y} = x \right)$

iv. $\exists x \forall y (y + 1 = x)$

4. Suppose you are given three sets defined as below:

$$A = \{x \in \mathbb{Z}^+ \mid x \text{ is a multiple of 2 and } x \leq 12\},$$

$$B = \{1, 2, 4, 6, 8, 10, 12\}, \quad C = \{2, 3, 5, 7\}$$

- (a) List all elements of set A . [1]

- (b) Find $(B - A) \cup C$ and determine its cardinality. [1+1]

- (c) Find $(A \cup C) \cap (B - C)$. [2]

- (d) Find the power set of $(A - B)$. [1]

- (e) Find $|(C - B) \times (A \cap C)|$. [2]

5. (a) Prove the following statement using an appropriate proof technique: "If n is an integer and n is even, then $10n^2 + 8n + 2$ is even." [3]

- (b) Prove the following statement using an appropriate proof technique: "The sum of an irrational and a rational number is irrational" [3]